

# TMIRI

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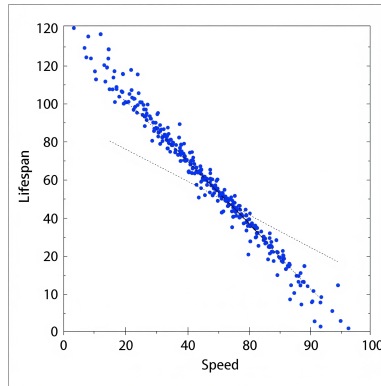
## Epistemology Test October 2025. Duration: 90'

You must be **brief** and **precise** in your answers.

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1. (1.5 point) Explain what is the objective of the deductive step of Scientific Method and why it makes sense to also call it **failed falsification** step.
2. (2 point, wrong answers subtract .5) According to Daniel Kahnemann's model of the human mind, Say if the following statements are right or wrong.
  - Our brain has two ways of thinking: fast and slow. Fast thinking is intuitive and inaccurate and slow thinking is deliberative and accurate.
  - The challenge of education is to train our brain to turn off fast thinking and improve slow thinking, so our brain only does sound reasoning.
  - Only experts reasoning about their field of expertise can avoid cognitive biases and therefore be sure that their reasoning is sound
  - Fast thinking is more tiring than slow thinking because it uses more body energy.
3. (2 points) Consider the following paragraph and explain the type(s) of reasoning that appear.

A biologist discovers a new bacteria. She records data from many samples. In particular, when she plots the speed of their Brownian motion vs their lifespan she gets the plot below. From that she concludes that the lifespan of this bacteria is inversely proportional to its speed. She writes in her notebook that probably Brownian motion damages the membrane anticipating the dead. In a second experiment she wants to inoculate this bacteria to mice brains and she needs the bacteria to live as much as possible inside the brain. To achieve this goal, she selects for inoculation only the slowest bacteria.



4. (1.5 points) Explain why observational research is not appropriate to prove causality
5. (1.5 points)
  - What is systematic observation?
  - Why it is important in Science?
  - Who was the first person realizing its importance?
6. (1.5 points) Consider the abstract of a Scientific paper that appears below. Label the type of research that is reported according to the different classifications that we have seen in class

This paper presents a study of software development team dynamics and their impact on code quality over a two-year period. We tracked 50 development teams, comprising a total of 350 engineers, from 10 different companies. We focused on collecting data on team communication patterns, knowledge sharing practices, and developer turnover without introducing any experimental interventions. Data was collected through a combination of automated analysis of version control systems (e.g., Git) to measure code churn and complexity, and anonymized surveys to assess team cohesion and developer satisfaction. Our findings reveal a strong correlation between stable team compositions and higher code quality, measured by a reduction in reported bugs and technical debt. Conversely, teams experiencing frequent developer turnover and fragmented communication channels exhibited a significant increase in software defects over time. The results suggest that fostering team stability and effective knowledge transfer is a critical, though often overlooked, factor in maintaining long-term software health. The study provides a foundational understanding of the socio-technical factors that drive software quality and offers practical insights for engineering managers.